

School of Computer Science and Engineering

Digital Assignment – I

SLOT :A1

Programme Name & Branch : B.Tech - CSE
Course Name & code : BCSE204L – Design and Analysis of Algorithms
Class Number (s) : VL2023240500925
VTOP submission date : 25- FEB-2024

GREEDY TECHNIQUE

1. Find the number of non-conflicting activities that can be executed by a single person for the given time frame.

Set = {A1, A2, A3, A4, A5, A6, A7, A8, A9, A10}

Start Time = {3, 2, 4, 8, 7, 9, 1, 5, 6, 10}

End Time = {4, 3, 4, 10, 9, 10, 4, 7, 7, 11}

2. Find the Huffman code for given character set and their frequency. Using the Huffman code decode the message “0110111001010”

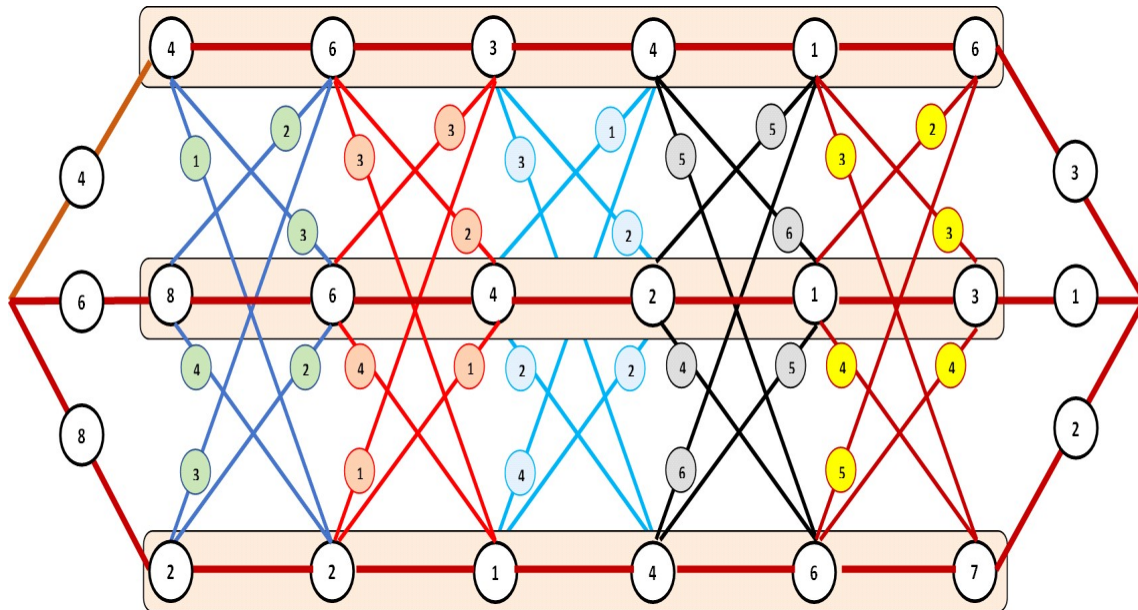
A	B	C	D	E	F
5	25	7	5	4	12

DIVIDE AND CONQUER

3. Find the maximum subarray sum for given set $S = \{-8, 7, 6, -5, -2, 4, 3, -5\}$
4. Perform Karatsuba Multiplication for given integers 14378, 679

DYNAMIC PROGRAMMING

5. Consider a factory with three assemble line each with 6 stations. The Matrix A represent Execution time of each station in the assemble line. The matrix T represents the transfer the time one station to another station of another assemble line. E represent the entry time and X represent the exit time from assemble line.



6. Find in which order the given matrix chain multiplication would have performed (A1.A2.A3.A4.A5.A6).

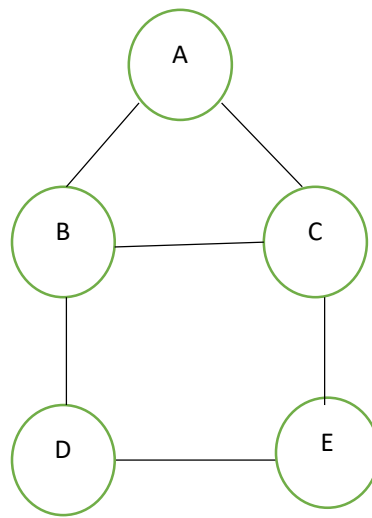
Matrix	A1	A2	A3	A4	A5	A6
Dimensions	10X4	4X2	2X3	3X5	5X6	6X5

7. Find the length of longest common subsequence and all possible subsequence for given string "ATGGATBBT" "GTABTABAT"
8. Find the optimal solution for given 0/1 Knapsack Problem with $W=9$.
Profit = {6, 5, 2, 4, 3} Weight = {2, 4, 5, 1, 3}.
9. Solve Travelling Salesman Problem using Dynamic programming for given graph G with vertices A, B, C, D, E.

$$G = \begin{bmatrix} 0 & 2 & 12 & 4 & 8 \\ 2 & 0 & 5 & 7 & 3 \\ 12 & 5 & 0 & 10 & 1 \\ 4 & 7 & 10 & 0 & 4 \\ 8 & 3 & 1 & 4 & 0 \end{bmatrix}$$

BACKTRACKING

10. Find subset sum for $n=5$, $W = \{3, 5, 2, 3, 4\}$, Sum = 8.
11. Solve Graph coloring for given graph find how many colors is needed?





BRANCH AND BOUND

12. Solve the job sequence Problem using LC- BB and find variable length solution.

	Job			
	1	2	3	4
Profit	10	6	5	3
Deadline	2	2	3	4
Time	1	2	4	2

13. Find the combinational optimized solution for given 0/1 Knapsack Problem with W=8.

Profit = {3, 4, 5, 6} Weight = {3, 3, 2, 5}.

Q1

Set = $\{A1, A2, A3, A4, A5, A6, A7, A8, A9, A10\}$ Start = $\{3, 2, 4, 8, 7, 9, 1, 5, 6, 10\}$ end = $\{4, 3, 4, 10, 9, 10, 4, 7, 7, 11\}$

Arranging activities in increasing order of end time

Activity	A2	A1	A3	A7	A8	A9	A5	A4	A6	A10
start	2	3	4	1	5	6	7	8	9	10
end	3	4	4	4	7	7	9	10	10	11

Now, we select activity whose start time $\geq$ finish time of selected activity

1) A2 is selected first

Selected (A2)

2) A1 is selected because start time of A1 = 3 = end time of A2

Selected (A2, A1)

3) A3 is selected because start time of A3 = 4 = end time of A1

Selected (A2, A1, A3)

4) A7 is rejected because start time of A7 < end time of A3

5) A8 is selected because start time of A8 = 5 > end time of A3 = 4

Selected (A2, A1, A3, A8)

6) A9 is rejected because start time of A9 < end time of A8

7) A5 is selected because start time of A5 = 7 = end time of A8

Selected (A2, A1, A3, A8, A5)

8) A4 is rejected because start time of A4 < end time of A5

6) A6 is selected because start time of A6 = 9 = end time of A5

Selected (A2, A1, A3, A8, A5, A6)

7) A10 is selected because start time of A10 = 10 = end time of A6

Selected (A2, A1, A3, A8, A5, A6, A10)

~~Required~~

Activity	A2	A1	A3	A8	A5	A6	A10
start	2	3	4	5	7	9	10
end	3	4	4	7	9	10	11

Total number of non-conflicting activities that can be executed by a single person is 7

21BCE0358

Q2 Arranging in increasing order of frequency, we get;

character frequency

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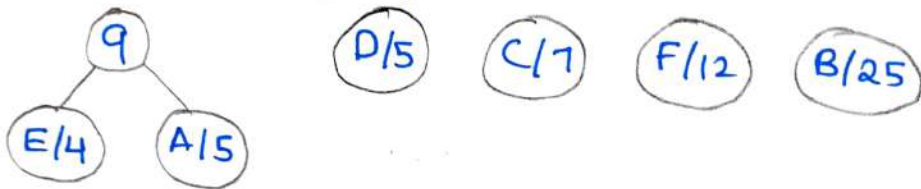
E	4
A	5
D	5
C	7
F	12
B	25

Step 1 -

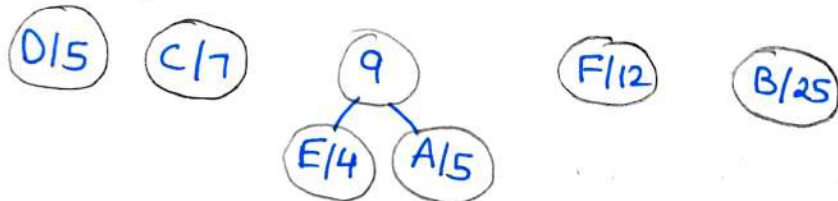


Step 2 -

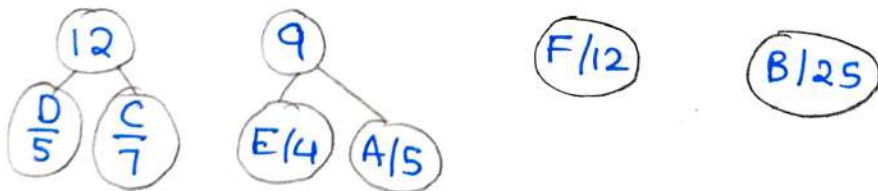
pairing lowest frequencies



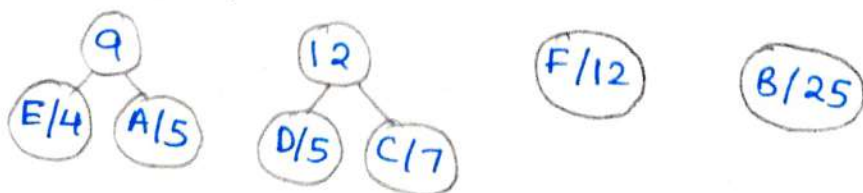
Rearranging,



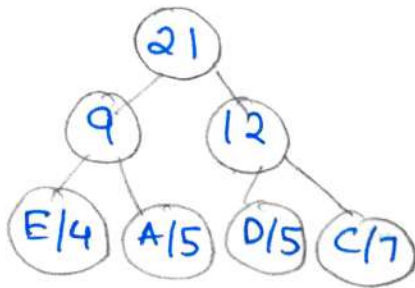
Step 3 -



Rearranging,



Step 4 -

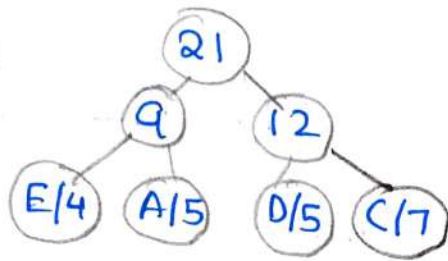


F/12

B/25

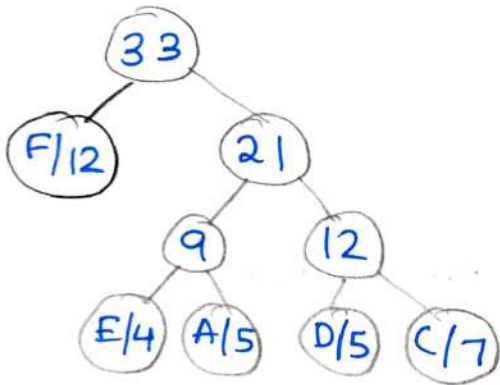
Rearranging,

F/12



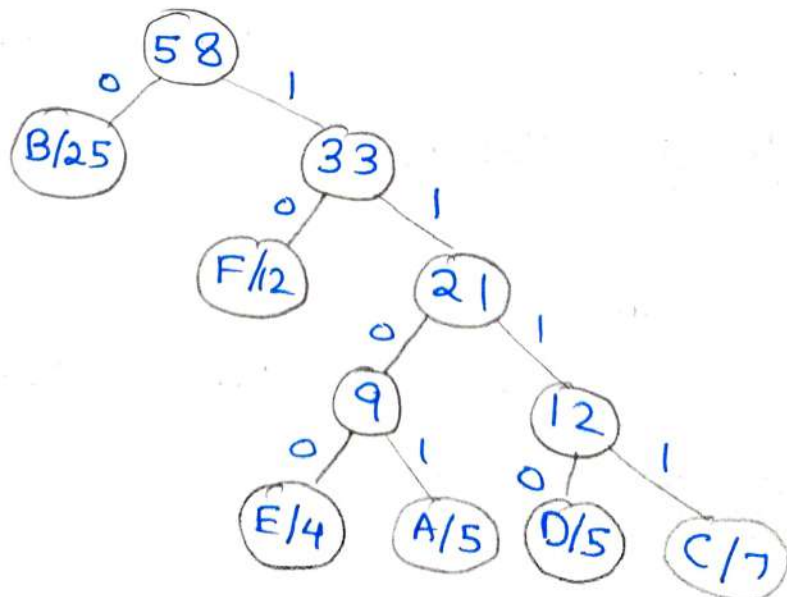
B/25

Step 5 -



B/25

Step 6 -



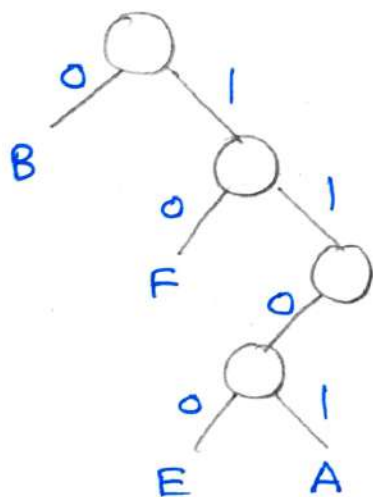
Codes -

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Character	code
A	1101
B	0
C	1111
D	1110
E	1100
F	10

Huffman decoding -

code: 0110111001010

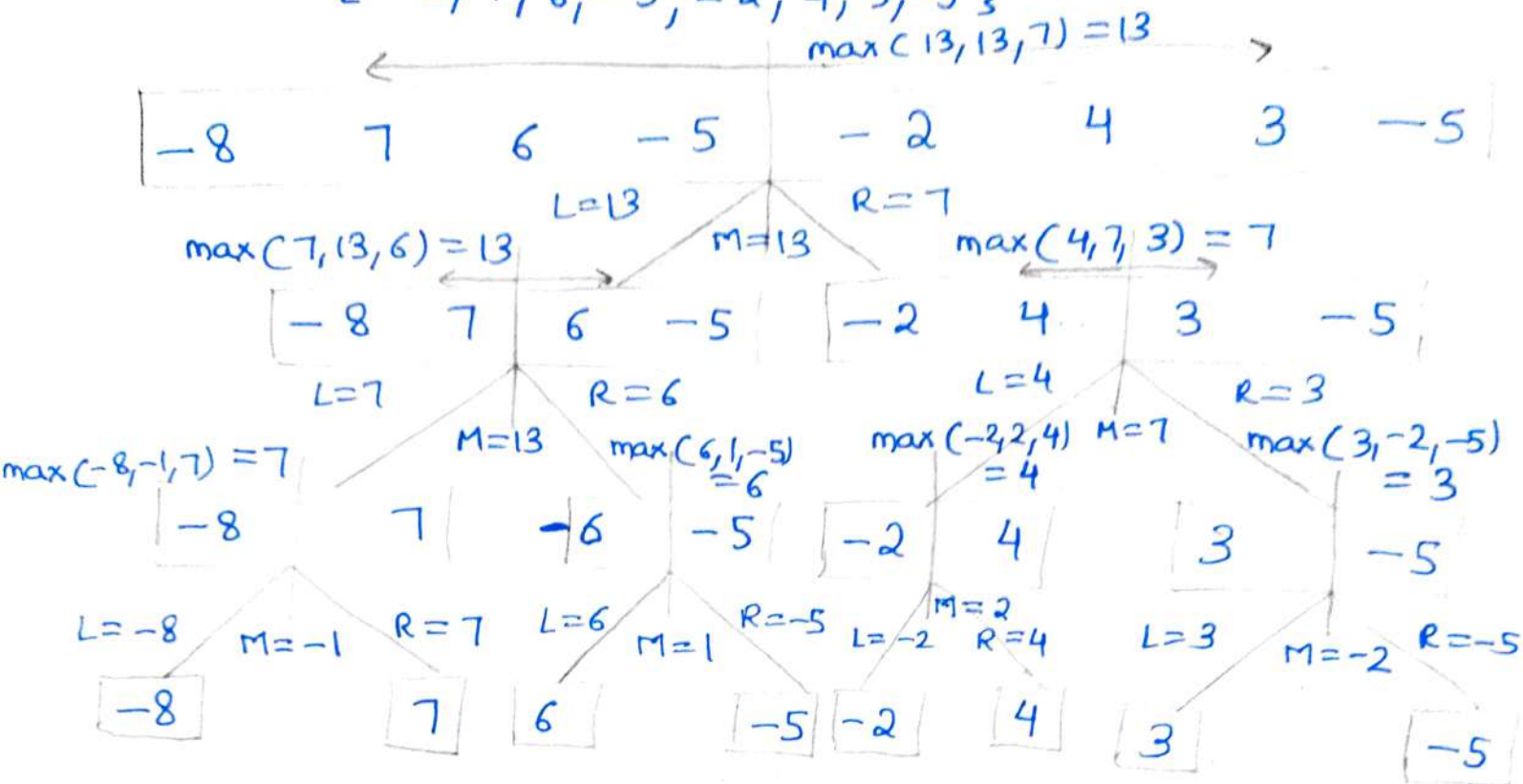


message is - B A E F F

Q3

21BCE0358

$$S = \{-8, 7, 6, -5, -2, 4, 3, -5\}$$



Maximum Subarray Sum = 13

Array index : $[1-2], [1-6]$

Array elements - $[7, 6], [7, 6, -5, -2, 4, 3]$

Q4

$$14378 \times 679$$

21BCE0358

$$n = \max \{ \text{size_base10}(14378), \text{size_base10}(679) \}$$

$$= \max \{ 5, 3 \} = 5$$

$$m = n/2 = 5/2 = 2$$

$$14378 = 143 \times 10^2 + 78$$

$$679 = 6 \times 10^2 + 79$$

$$x_1 = 143 \quad x_2 = 78$$

$$y_1 = 6 \quad y_2 = 79$$

$$a = x_1 y_1 = 143 \times 6 = 858$$

$$c = x_2 y_2 = \underbrace{78 \times 79}_{\rightarrow \textcircled{A}}$$

$$\begin{aligned} b &= (x_1 + x_2)(y_1 + y_2) - a - c \\ &= (143 + 78)(6 + 79) - 858 - 78 \times 79 \\ &= \underbrace{221 \times 85}_{\rightarrow \textcircled{B}} - 858 - \textcircled{A} \end{aligned}$$

solving for $\textcircled{A}$,

$$78 \times 79$$

$$\cancel{78} \quad n = \max \{ 2, 2 \} = 2$$

$$m = n/2 = 2/2 = 1$$

$$78 = 7 \times 10^1 + 8$$

$$79 = 7 \times 10^1 + 9$$

$$x_1 = 7 \quad x_2 = 8$$

$$y_1 = 7 \quad y_2 = 9$$

$$a = x_1 y_1 = 49 \quad ; \quad c = x_2 y_2 = 72$$

$$\begin{aligned}
 b &= (x_1 + x_2)(y_1 + y_2) - a - c \\
 &= (7+8)(7+9) - 49 - 72 \\
 &= 15 \times 16 - 49 - 72 \\
 &\quad \swarrow \searrow \rightarrow \textcircled{C}
 \end{aligned}$$

Solving for $\textcircled{C}$ - 15×16

$$n = \max\{2, 2\} = 2$$

$$m = n/2 = 1$$

$$15 = 1 \times 10^1 + 5$$

$$16 = 1 \times 10^1 + 6$$

$$x_1 = 1, \quad x_2 = 5$$

$$y_1 = 1, \quad y_2 = 6$$

$$a = x_1 y_1 = 1$$

$$c = x_2 y_2 = 30$$

$$\begin{aligned}
 b &= (x_1 + x_2)(y_1 + y_2) - a - c \\
 &= (1+5)(1+6) - 1 - 30 \\
 &= 6 \times 7 - 31 = 42 - 31 = 11
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow 15 \times 16 &= a \times 10^2 + b \times 10^1 + c = 1 \times 10^2 + 11 \times 10^1 + 30 = 100 + 110 + 30 \\
 &= 240
 \end{aligned}$$

Substituting back for $\textcircled{C}$ in $\textcircled{A}$

$$\begin{aligned}
 b &= 15 \times 16 - 49 - 72 \\
 &= 240 - 49 - 72 = 119
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow 78 \times 79 &= a \times 10^2 + b \times 10^1 + c = 49 \times 10^2 + 119 \times 10^1 + 72 \\
 &= 4900 + 1190 + 72 \\
 &= 6162
 \end{aligned}$$

Solving subproblem (B)

21BCE0358

$$221 \times 85$$

$$n = \max\{3, 2\} = 3$$

$$m = 3/2 = 1$$

$$221 = 22 \times 10^1 + 1$$

$$85 = 8 \times 10^1 + 5$$

$$x_1 = 22 ; x_2 = 1$$

$$y_1 = 8 ; y_2 = 5$$

$$a = x_1 y_1 = 22 \times 8 = 176$$

$$c = x_2 y_2 = 1 \times 5 = 5$$

$$b = (x_1 + x_2)(y_1 + y_2) - a - c$$

$$= (22 + 1)(8 + 5) - 176 - 5$$

$$= 23 \times 13 - 176 - 5$$

$$\searrow \text{ (D)}$$

Solving subproblem (D)

$$23 \times 13$$

$$n = \max\{2, 2\} = 2$$

$$m = 2/2 = 1$$

$$23 = 2 \times 10^1 + 3$$

$$13 = 1 \times 10^1 + 3$$

$$x_1 = 2 ; x_2 = 3$$

$$y_1 = 1 ; y_2 = 3$$

$$a = x_1 y_1 = 2$$

21BCE0358

$$c = x_2 y_2 = 9$$

$$\begin{aligned} b &= (x_1 + x_2)(y_1 + y_2) - a - c \\ &= 5 \times 4 - 2 - 9 \\ &= 20 - 11 = 9 \end{aligned}$$

$$\begin{aligned} 23 \times 13 &= a \times 10^2 + b \times 10^1 + c = 200 + 90 + 9 \\ &= 299 \end{aligned}$$

Substituting back in (B)

$$\begin{aligned} b &= 23 \times 13 - 176 - 5 \\ &= 299 - 181 = 118 \end{aligned}$$

~~Substituting~~ $221 \times 85 = a \times 10^2 + b \times 10^1 + c$

$$\begin{aligned} &= 176 \times 10^2 + 118 \times 10 + 5 \\ &= 18785 \end{aligned}$$

Substituting back in top level problem

$$\begin{aligned} b &= 221 \times 85 - 858 - \textcircled{A} \\ &= 18785 - 858 - 6162 \\ &= 11765 \end{aligned}$$

$$\begin{aligned} \Rightarrow 14378 \times 679 &= a \times 10^4 + b \times 10^2 + c \\ &= 858 \times 10^4 + 11765 \times 10^2 + 6162 \\ &= 9762662 \end{aligned}$$

Q5 Base case, $j=1$ for lines $i=1, 2, 3$

21BCE0358

$$f_{1[1]} = c_1 + A_{11}$$

$$= 3 + 3 = 6$$

$$f_{2[1]} = c_2 + A_{21}$$

$$= 1 + 1 = 2$$

$$f_{3[1]} = c_3 + A_{31}$$

$$= 2 + 4 = 6$$

	1	2	3	4	5	6
$f_{1[j]}$	6	$\min(7, 5, 10) = 5$	$\min(8, 10, 12) = 8$	$\min(11, 11, 17) = 11$	$\min(15, 19, 21) = 15$	$\min(16, 17, 19) = 16$
$f_{2[j]}$	2	$\min(11, 4, 10) = 4$	$\min(10, 7, 12) = 7$	$\min(13, 10, 17) = 10$	$\min(21, 14, 20) = 14$	$\min(23, 19, 22) = 19$
$f_{3[j]}$	6	$\min(9, 8, 8) = 8$	$\min(12, 12, 12) = 12$	$\min(13, 11, 14) = 11$	$\min(18, 16, 13) = 13$	$\min(20, 20, 15) = 15$

	1	2	3	4	5	6
L_1	1	2	1	1	1	1
L_2	2	2	2	2	2	2
L_3	3	2	3	2	3	3

$$f^* = \min(16 + 8, 19 + 4, 15 + 6)$$

$$= \min(24, 23, 21) = 21$$

$$L^* = 3$$

Minimum Cost = 21, exit from line 3

Q6

21BCE0359

matrix

A1

A2

A3

A4

A5

A6

dimensions

10x4

4x2

2x3

3x5

5x6

6x5

 d_0 d_1 d_2 d_3 d_4 d_5 d_6

	1	2	3	4	5	6		1	2	3	4	5	6
1	0	80	140	210	290	330	1		1	2	2	2	2
2		0	24	70	138	190	2			2	2	2	2
3			0	30	90	150	3				3	4	5
4				0	90	180	4					4	5
5					0	180	5						5
6						0	6						

for base case, $m[i, i+1] = d_{i-1} \times d_i \times d_{i+1}$

this is the least cost for multiplying i th and $(i+1)$ th matrix

$$m[1, 1] = 0$$

$$m[4, 4] = 0$$

$$m[2, 2] = 0$$

$$m[5, 5] = 0$$

$$m[3, 3] = 0$$

$$m[6, 6] = 0$$

$$m[1, 2] = d_0 \times d_1 \times d_2 = 10 \times 4 \times 2 = 80$$

$$m[2, 3] = d_1 \times d_2 \times d_3 = 4 \times 2 \times 3 = 24$$

$$m[3, 4] = d_2 \times d_3 \times d_4 = 2 \times 3 \times 5 = 30$$

$$m[4, 5] = d_3 \times d_4 \times d_5 = 3 \times 5 \times 6 = 90$$

$$m[5, 6] = d_4 \times d_5 \times d_6 = 5 \times 6 \times 5 = 150$$

general formula -

multiplying i th to j th is

$$m[i, j] = \min [m[i, k] + m[k+1, j] + d_{i-1} \times d_k \times d_j] \text{ where } k = i \text{ to } j-1$$

$$\begin{aligned} m[1, 3] &= \min [m[1, 1] + m[2, 3] + d_0 \times d_1 \times d_3, \\ &\quad m[1, 2] + m[3, 3] + d_0 d_2 d_3] \\ &= \min [0 + 24 + 10 \times 4 \times 3, 80 + 0 + 10 \times 2 \times 3] \\ &= \min [144, 140] = 140 \end{aligned}$$

$$\begin{aligned} m[2, 4] &= \min [m[2, 2] + m[3, 4] + d_1 \cdot d_2 \cdot d_4, \\ &\quad m[2, 3] + m[4, 4] + d_1 \cdot d_3 \cdot d_4] \\ &= \min [0 + 30 + 4 \times 2 \times 5, 24 + 0 + 4 \times 3 \times 5] \\ &= \min [70, 84] = 70 \end{aligned}$$

$$\begin{aligned} m[3, 5] &= \min [m[3, 3] + m[4, 5] + d_2 \cdot d_3 \cdot d_5, \\ &\quad m[3, 4] + m[5, 5] + d_2 \cdot d_4 \cdot d_5] \\ &= \min [0 + 90 + 2 \times 3 \times 6, 30 + 0 + 2 \times 5 \times 6] \\ &= \min [126, 90] = 90 \end{aligned}$$

$$\begin{aligned} m[4, 6] &= \min [m[4, 4] + m[5, 6] + d_3 \cdot d_4 \cdot d_6, \\ &\quad m[4, 5] + m[6, 6] + d_3 \cdot d_5 \cdot d_6] \\ &= \min [0 + 180 + 3 \times 5 \times 5, 90 + 0 + 3 \times 6 \times 5] \\ &= \min [255, 180] = 180 \end{aligned}$$

$$m[1,4] = \min [m[1,1] + m[2,4] + d_0 \cdot d_1 \cdot d_4, \\ m[1,2] + m[3,4] + d_0 \cdot d_2 \cdot d_4, \\ m[1,3] + m[4,4] + d_0 \cdot d_3 \cdot d_4]$$

$$= \min [0 + 70 + 10 \times 4 \times 5, 80 + 30 + 10 \times 2 \times 5, 140 + 0 + 10 \times 3 \times 5]$$

$$= \min [270, 210, 290] = 210$$

$$m[2,5] = \min [m[2,2] + m[3,5] + d_1 \cdot d_2 \cdot d_5, \\ m[2,3] + m[4,5] + d_1 \cdot d_3 \cdot d_5, \\ m[2,4] + m[5,5] + d_1 \cdot d_4 \cdot d_5]$$

$$= \min [0 + 90 + 4 \times 2 \times 6, 24 + 90 + 4 \times 3 \times 6, 70 + 0 + 4 \times 5 \times 6]$$

$$= \min [138, 186, 190] = 138$$

$$m[3,6] = \min [m[3,3] + m[4,6] + d_2 \cdot d_3 \cdot d_6, \\ m[3,4] + m[5,6] + d_2 \cdot d_4 \cdot d_6, \\ m[3,5] + m[6,6] + d_2 \cdot d_5 \cdot d_6]$$

$$= \min [0 + 180 + 2 \times 3 \times 5, 30 + 150 + 2 \times 5 \times 5, 90 + 0 + 2 \times 6 \times 5]$$

$$= \min [210, 230, 150] = 150$$

$$m[1,5] = \min [m[1,1] + m[2,5] + d_0 \cdot d_1 \cdot d_5, \\ m[1,2] + m[3,5] + d_0 \cdot d_2 \cdot d_5, \\ m[1,3] + m[4,5] + d_0 \cdot d_3 \cdot d_5, m[1,4] + m[5,5] + \\ d_0 \cdot d_4 \cdot d_5]$$

$$= \min [0 + 138 + 10 \times 4 \times 6, 80 + 90 + 10 \times 2 \times 6, 140 + 90 + 10 \times 3 \times 6, \\ 210 + 0 + 10 \times 5 \times 6]$$

$$= \min [378, 290, 410, 510] = 290$$

$$\begin{aligned}
 m[2,6] &= \min[m[2,2] + m[3,6] + d_1 \cdot d_2 \cdot d_6, \\
 &\quad m[2,3] + m[4,6] + d_1 \cdot d_3 \cdot d_6, \\
 &\quad m[2,4] + m[5,6] + d_1 \cdot d_4 \cdot d_6, \\
 &\quad m[2,5] + m[6,6] + d_1 \cdot d_5 \cdot d_6]
 \end{aligned}$$

$$= \min[0 + 150 + 4 \times 2 \times 5, 24 + 180 + 4 \times 3 \times 5, 70 + 150 + 4 \times 5 \times 5, 138 + 0 + 4 \times 6 \times 5]$$

$$= \min[190, 264, 320, 258] = 190$$

$$\begin{aligned}
 m[1,6] &= \min[m[1,1] + m[2,6] + d_0 \cdot d_1 \cdot d_6, \\
 &\quad m[1,2] + m[3,6] + d_0 \cdot d_2 \cdot d_6, \\
 &\quad m[1,3] + m[4,6] + d_0 \cdot d_3 \cdot d_6, \\
 &\quad m[1,4] + m[5,6] + d_0 \cdot d_4 \cdot d_6, \\
 &\quad m[1,5] + m[6,6] + d_0 \cdot d_5 \cdot d_6]
 \end{aligned}$$

$$= \min[0 + 190 + 10 \times 4 \times 5, 80 + 150 + 10 \times 2 \times 5, 140 + 180 + 10 \times 3 \times 5, 210 + 150 + 10 \times 5 \times 5, 290 + 0 + 10 \times 6 \times 5]$$

$$= \min[390, 330, 470, 610, 590] = 330$$

Minimum cost is 330

$((A_1 \cdot A_2) \cdot ((A_3 \cdot A_4) \cdot A_5) \cdot A_6)$ is the order of multiplication

Q7

R1BCE0358

"	G	T	A	B	T	A	B	A	T
"	0	0	0	0	0	0	0	0	0
A	0	0	0	1	1	1	1	1	1
T	0	0	1	1	1	2	2	2	2
G	0	1	1	1	1	2	2	2	2
G	0	1	1	1	1	2	2	2	2
A	0	1	1	2	2	3	3	3	3
T	0	1	2	2	2	3	3	3	4
B	0	1	2	2	3	3	4	4	4
B	0	1	2	2	3	3	4	4	4
T	0	1	2	2	3	4	4	4	5

longest subsequence will be of length 5

longest subsequences -

GABBT, GATBT, TATBT, TABBT, GTBBT,
ATABT

Q8

21BCE0358

item	profit	weight
1	4	1
2	6	2
3	3	3
4	5	4
5	2	5

$$w=9$$

j

			0	1	2	3	4	5	6	7	8	9
P_i	w_i	0	0	0	0	0	0	0	0	0	0	0
4	1	1	0	4	4	4	4	4	4	4	4	4
6	2	2	0	4	6	10	10	10	10	10	10	10
3	3	3	0	4	6	10	10	10	10	10	10	10
5	4	4	0	4	6	10	10	10	13	15	15	15
2	5	5	0	4	6	10	10	10	13	15	15	15

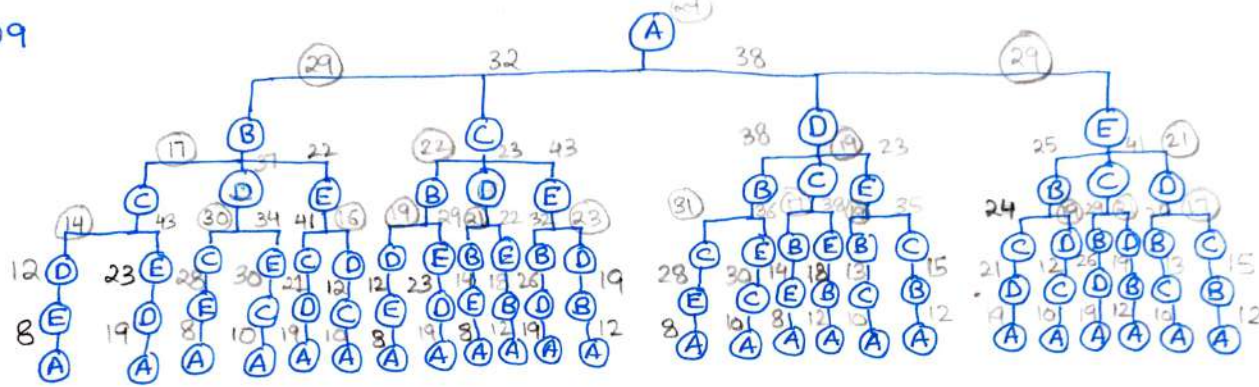
$$T(i, j) = \begin{cases} T(i-1, j) & \text{if } j < w_i \\ \max \{ T(i-1, j), v_i + T(j - w_i) \} & \text{otherwise} \end{cases}$$

maximum profit = 15

$$\text{vector} = \{1, 1, 0, 1, 0\}$$

Q9

21BCE0358



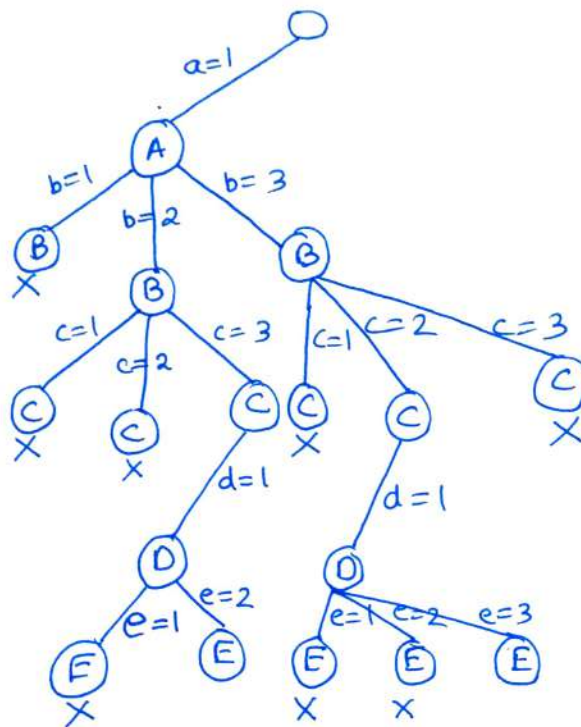
	A	B	C	D	E
A	0	12	10	19	8
B	12	0	3	7	6
C	10	3	0	2	20
D	19	7	2	0	4
E	8	6	20	4	0

path- ① A B C D E A

② A E D C B A

2 BCE0358

Solution set : $\{1, 1, 0, 0, 0\}, \{1, 0, 1, 1, 0\}, \{0, 1, 0, 1, 0\}$



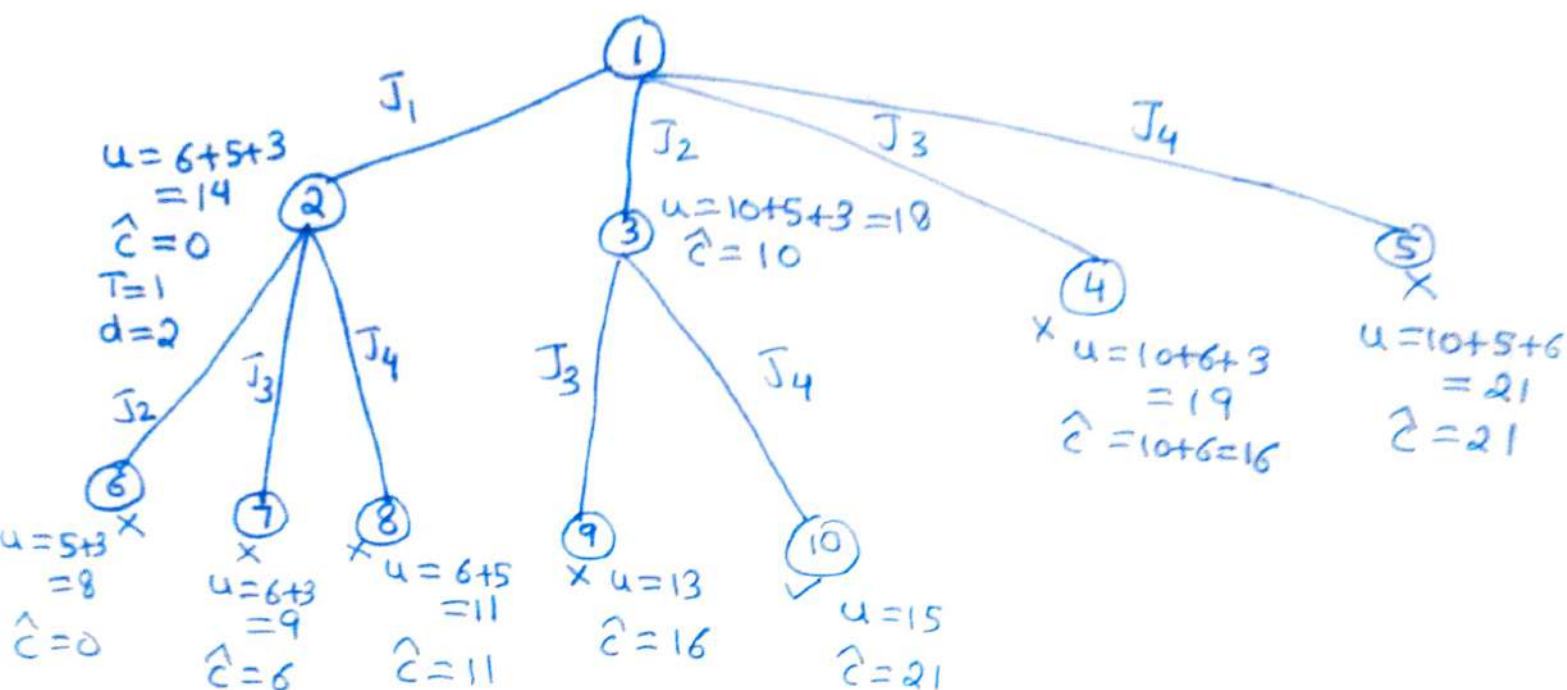
from the above tree, the possible solutions are-

s.no	A	B	C	D	E
1.	1	2	3	1	2
2.	1	3	2	1	3

∴ minimum number of colors required is 3

Q12

$$u = \phi \quad 1/4 \quad 1/8 \quad 1/4 \quad 21$$



optimal solution is J_2, J_4

Q13

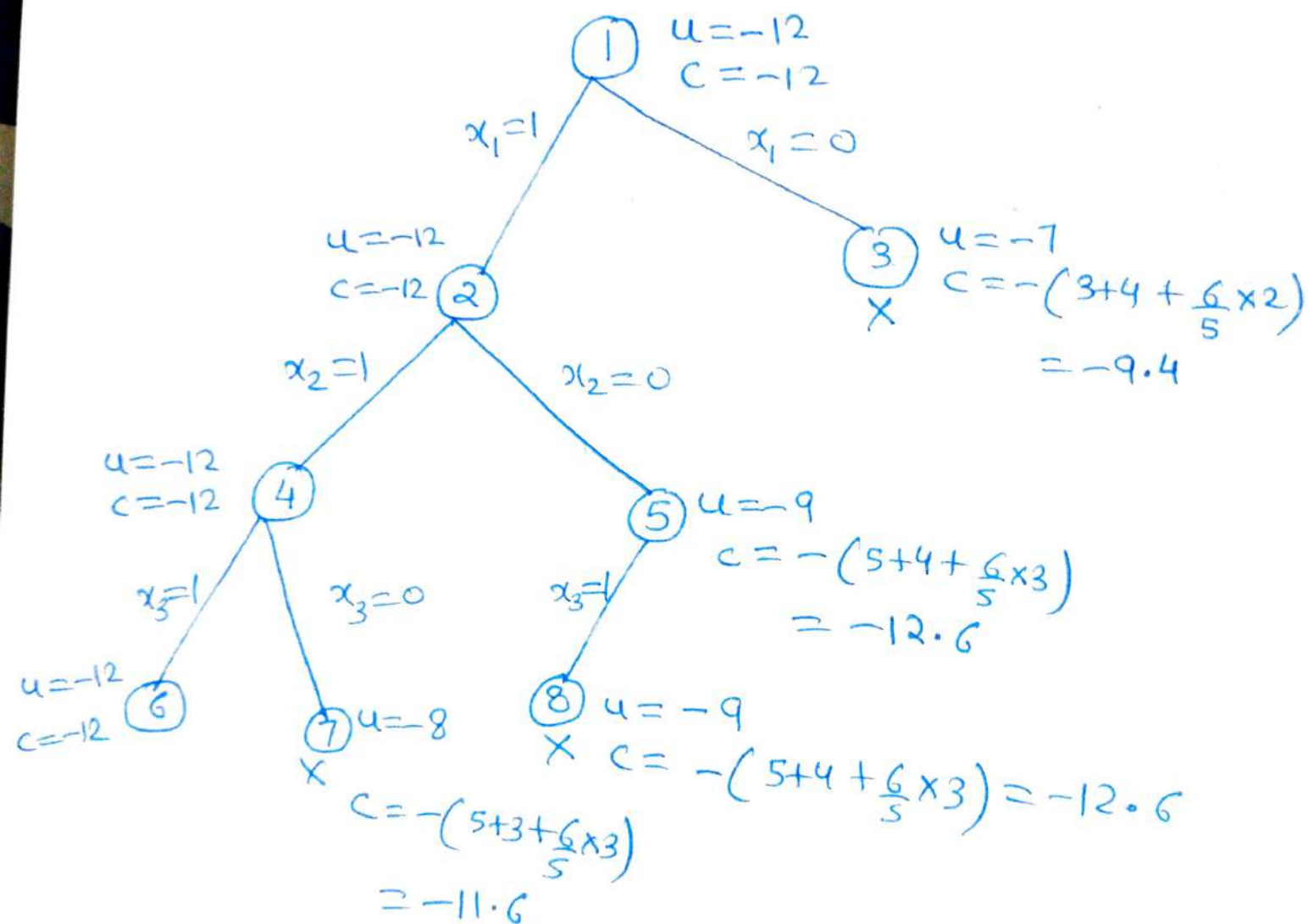
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profit	5	3	4	6
weight	2	3	3	5

$$W = 8$$

$$C = - \left(5 + 3 + 4 + \frac{6}{5} \times 0 \right) = -12$$

$$u = -\phi -12$$



Solution set : $\{1, 1, 1, 0\}$

profit = 12